Day : 01

* The OSI Model

- What is the OSI Model?
  - Open System Interconnection Reference Model

- It's a guide (thus the term "model")
  - Don't get wrapped up in the details

- This is not the OSI protocol suite
  - Most of the OSI protocol suite didn't catch on

- There are unique protocols at every layer

- You'll refer to this model for the rest of your career

- All people Seem To Need Data Processing

Layer 7
Application
Layer 6
Presentation
Layer 5
Session
Layer 4
Transport
Layer 3
Network
Layer 2
Data Link
Layer 1
Physical

Page No.
Date

* Layer 1 – Physical Layer

- The physics of the network
  - Signaling, cabling, connectors
  - This layer isn't about protocols

- You have a physical layer problem:
  - Fix your cabling, punch-downs, etc.
  - Run loopback tests, test / replace cables, swap adaptor cards.

* Layer 2 – Data Link Layer

- The basic network "language"
  - The foundations of communication at the data link layer

- Data Link Control (DLC) protocols
  - MAC (Media Access Control) address on Ethernet

- The "switching" layer

* Layer 3 – Network Layer

- The "routing" layer

- Internet Protocol (IP)

- Fragments frames to traverse different networks

* Layer 4 – Transport Layer

- The "post office" layer
  - Parcels and letters
- TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)

* Layer 5 – Session Layer

- Communication management between devices
  - Start, Stop, Restart
- Control protocols, tunneling protocols

* Layer 6 – Presentation Layer

- Character encoding
- Application encryption
- Often combined with the Application Layer

* Layer 7 – Application Layer

- The layer we see
- HTTP, FTP, DNS, POP3

* Cyber attacks at every OSI Layer

- Physical Layer
  - Cable taping, hardware theft, signal jamming

Page No.
Date

- Data Link Layer
  — MAC spoofing, ARP spoofing, VLAN hopping, Switch port stealing

- Network Layer
  — IP spoofing, ICMP flooding, smurf attack, Routing attack

- Transport Layer
  — TCP SYN flood, UDP floods, ~~Se~~ TCP reset attack

- Session Layer
  — Session Hijacking, Session fixation

- Presentation Layer
  — SSL / TLS attacks, Encryption attack, Data format exploitation

- Application Layer
  — SQL Injection, Cross-Site Scripting (XSS), Malware injection, Phishing.